



Fig. 1: Ground based air defence against aerial targets



Fig. 2: Airborne defence of aircraft against shoulder-fired missiles



Fig. 3: Laser weapons for ship defence

system would draw about 400kW to 500kW of electrical power for a period of five seconds, which is the same as the electrical power consumed by a 100W bulb in seven hours.

Limitations

Laser based DEWs also have some limitations. Some of these include their line-of-sight dependence, requirement of finite dwell time, problems due to atmospheric attenuation and turbulence, and ineffectiveness against hardened structures.

Laser weapons require direct line-of-sight to engage a target. Their effectiveness is reduced or neutralised by the presence of an object or structure in front of the target that cannot be burned through.

Unlike projectile weapons that instantly destroy the target upon impact, laser weapons require a mini-

mum dwell time of the order of three to five seconds to deposit sufficient energy for target destruction.

Effectiveness of the laser weapon is adversely affected by atmospheric conditions. The laser beam suffers attenuation due to absorption and scattering by airborne particles and gas molecules, deterioration of beam quality in the form of deformation of the laser beam wave front and increase in the laser beam spot size at the target.

Laser weapons are not very effective against hardened structures. However, equipment such as antennae, sensors and external fuel stores mounted on these structures can be targeted effectively.

Operational scenario

The operational scenario of directed-energy laser weapons is broadly categorised as short- and medium-range tactical missions and long-range strategic missions. Some of the important application areas of tactical-class laser weapons include stand-off neutralisation of ordnances such as mines, unexploded ordnances and improvised explosive devices, ground based defence against rockets, artillery and mortars (Fig. 1), ground based capability to destroy unmanned aerial vehicles (UAVs) of the adversary, airborne defence of aircraft against man-portable air defence systems such as shoulder-fired surface-to-air missiles (Fig. 2), ship defence against maneuvering cruise missiles and tactical ballistic missiles (Fig. 3).

Stand-off neutralisation of ordnances requires laser power in the range of 1kW to 2kW for operational ranges up to 300 metres. Solid-state or fibre-laser sources operating around 1.0µm are used for the purpose. Avenger and Zeus of the USA and Israeli Thor are laser ordnance systems with comparable specifications.

Anti-UAV operations up to a range of eight to ten kilometres require about 100kW of laser power. Operational ranges of five to six kilo-

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Applications

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